CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging

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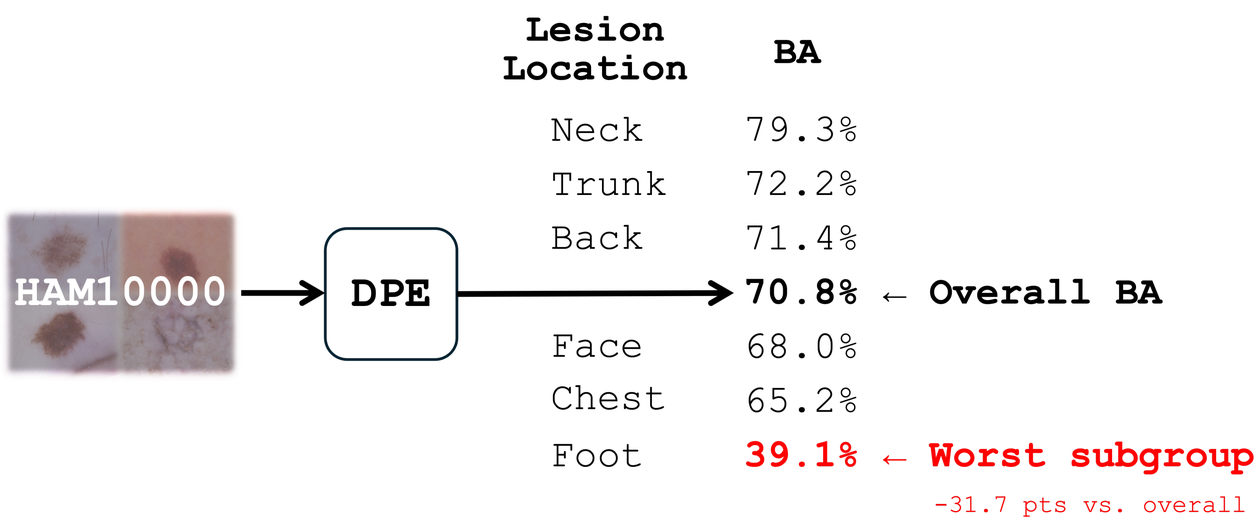
Paper

Slide

Motivation: Overall vs. Subgroup Failure

Medical imaging models may achieve strong overall performance while exhibiting substantial disparities across clinically meaningful subgroups. For example, on HAM10000, an image-only DPE model achieves 70.8% overall balanced accuracy, but only 39.1% for lesions located on the foot.

Traditional subgroup analysis relies on explicit demographic, anatomical, or acquisition metadata, which may be incomplete or unavailable in real-world deployment. Although image-derived proxies offer an alternative, their predictions can be uncertain, and not all proxy attributes are equally relevant to model failures.



Research Question

How can we perform reliable and interpretable subgroup analysis when explicit metadata are missing?

CAPRA Framework

Semantic Proxy Construction

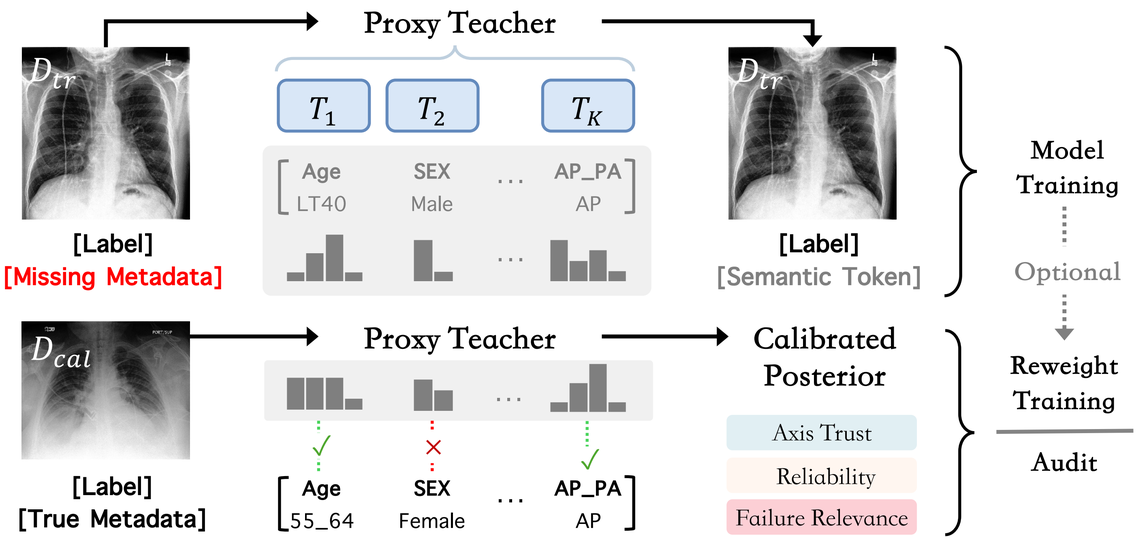
- Image-derived proxy teachers predict predefined semantic attributes, such as age, sex, and acquisition conditions. Structured semantic tokens are integrated with visual features to train a warm-start predictor.

Calibrated Proxy Posteriors

- A small metadata-labeled cohort enables posterior calibration through patient-level cross-fitting, temperature scaling, and confusion-matrix shrinkage. Calibrated probabilities preserve subgroup uncertainty, while entropy-based scores characterize sample reliability.

Trust- and Failure-Aware Axis Weighting

- CAPRA jointly considers proxy prediction quality (Axis Trust) and subgroup error concentration (Failure Relevance) to prioritize informative semantic axes.



Results

Semantic Alignment

CAPRA produces subgroup structures that are more closely aligned with predefined audit semantics than image-only and latent alternatives on NMI.

Source	BRSET			HAM10000			CheXpert		
	Pur./NMI ↑	Agr. (%) ↑	Gap (%) ↓	Pur./NMI ↑	Agr. (%) ↑	Gap (%) ↓	Pur./NMI ↑	Agr. (%) ↑	Gap (%) ↓
True metadata	**0.98** / **1.00**	**100.0**	**4.9**	**0.95** / **1.00**	**100.0**	**33.4**	**0.98** / **1.00**	**100.0**	**6.3**
Image-only	0.24 / 0.08	37.6	38.6	0.21 / 0.12	29.1	51.0	0.31 / 0.11	72.2	81.3
ExMap	0.08 / 0.06	37.4	36.3	0.23 / 0.11	27.8	51.4	**0.61** / 0.33	**84.2**	63.1
CAPRA	**0.81** / **0.60**	**83.3**	**19.0**	**0.68** / **0.68**	**81.7**	**10.8**	0.55 / **0.43**	70.5	**22.5**

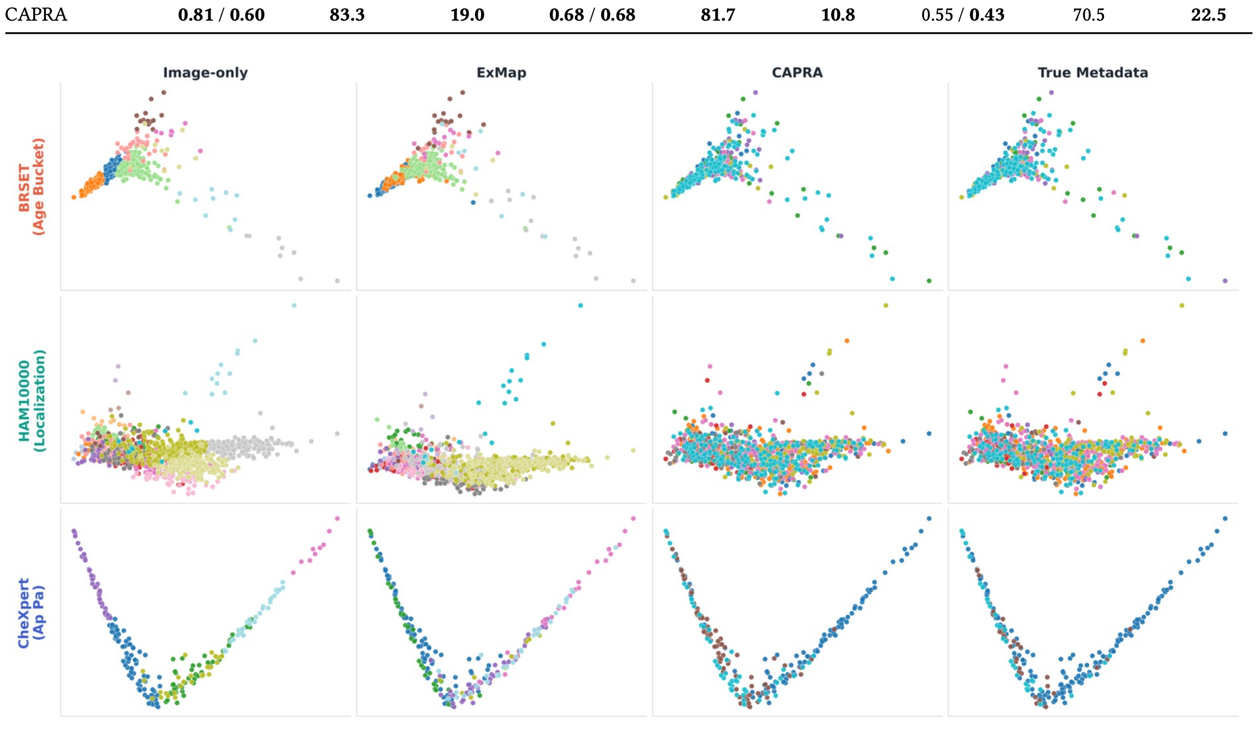
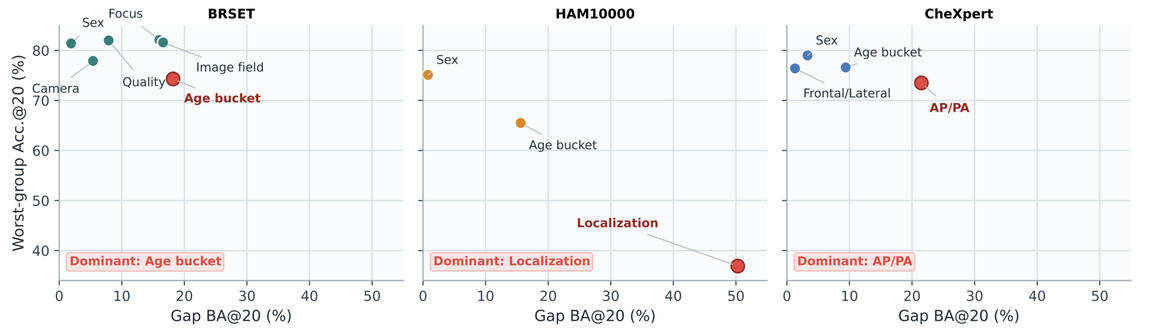


Figure 3: Aligned 2D subgroup partitions across BRSET, HAM10000, and CheXpert. Within each dataset, Image-only, ExMap, CAPRA, and True Metadata partitions are shown on the same embedding to visualize semantic alignment with the dominant explicit failure axis.

Subgroup Failure Patterns

The dominant failure axis varies across modalities, highlighting the need for domain-adaptive, semantically interpretable subgroup analysis.



Points farther right and lower indicate more severe explicit subgroup failure.

Downstream Gains

CAPRA provides reusable subgroup information that can improve worst-group robustness across different learners and imaging tasks.

Method	BRSET				HAM10000				CheXpert			
	BA (%)	Δ	WGA (%)	Δ	BA (%)	Δ	WGA (%)	Δ	BA (%)	Δ	WGA (%)	Δ
GroupDRO*	62.6±2.1	–	51.5±3.6	–	33.8±2.7	–	20.5±4.2	–	65.6±1.3	–	48.1±2.4	–
+ CAPRA	83.1±1.4	+20.5	69.9±2.0	+18.4	72.9±1.7	+39.1	33.2±3.3	+12.7	82.5±0.8	+16.9	61.4±1.9	+13.3
JTT	73.2±2.6	–	60.3±3.2	–	56.1±2.0	–	27.6±4.0	–	81.0±1.2	–	61.5±2.1	–
+ CAPRA	77.9±1.5	+4.7	73.2±2.3	+12.9	69.5±2.1	+13.4	41.0±2.7	+13.4	78.6±0.9	-2.4	66.9±1.7	+5.4
DFR	72.9±1.6	–	65.7±2.0	–	68.7±1.4	–	26.8±2.9	–	**89.5±0.7**	–	**77.7±1.3**	–
+ CAPRA	74.3±1.1	+1.4	70.4±1.9	+4.7	69.0±1.5	+0.3	28.0±2.1	+1.2	81.0±0.9	-8.5	74.5±1.2	-3.2
DPE	84.6±1.2	–	66.9±2.2	–	71.9±1.6	–	39.1±3.0	–	84.6±1.0	–	69.6±1.7	–
+ CAPRA	**86.0±0.7**	+1.4	**79.0±1.7**	+12.1	**74.5±1.1**	+2.6	**49.4±2.4**	+10.3	85.8±0.6	+1.2	71.5±1.8	+1.9
GSR	76.8±1.9	–	61.0±2.7	–	58.6±2.3	–	34.1±3.5	–	78.7±1.1	–	66.2±2.2	–
+ CAPRA	72.5±1.4	-4.3	64.4±2.6	+3.4	58.3±1.7	-0.3	41.1±2.9	+7.0	80.3±1.2	+1.6	68.3±1.6	+2.1